Teaching an LLM to Read Emotions in a Language None of Us Speaks

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Agenda

1 Introduction

2 Literature Review

3 Methodology

4 Implementation

5 Results & Discussion

6 Conclusion



Introduction

Detecting emotions in subtitles beyond English

- Our language: Croatian, resource band 3 of 5 (mid-low)
- An LLM's English skill says little about its Croatian
- Croatian has no human labels, only labels projected from English
- None of us reads Croatian, so we can't check the labels or the predictions ourselves

"You did what?"

anger · anticipation · disgust · fear
joy · sadness · **surprise** · trust

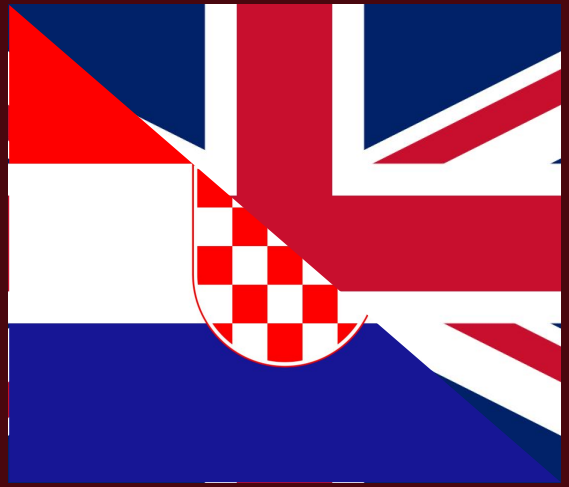
Multi-label: one line, no context, several emotions at once

Our three questions...



Prompting or fine-tuning?

Which works better on the same model?



Does English carry over?

Does training on English help on Croatian, or do we need Croatian data?



What do projected labels mean?

How do labels copied from English affect evaluation?

Literature Review

Crossing languages

- **XED:** Movie subtitles labelled with Plutchik's eight emotions.
  - One line can carry several emotions.
- **Task:** Eight yes/no decisions.
- **Annotation projection:** Subtitles are aligned by timing and labels copied across.
  - Aligned lines do not need to be translations of each other.
- Only English and Finnish were human-labelled for XED.
  - Croatian (+29 languages) inherited projected labels.
- **Cross-lingual transfer:** Train in one language, test in another.
  - We test English-only fine-tuning on Croatian.



Adapting LLMs

LoRA

- Freeze the model and train a small low-rank adapter.
- Trains at a fraction of a percent of its parameters.

QLoRA

- Like LoRA, but stores the frozen model in 4-bit.
- Memory for weights cut about fourfold.

Few-shot prompting

- A few solved examples in the prompt let a model do a new task.
- No fine-tuning, no changed weights.

Fragile prompts

- Few-shot results shift with wording and examples.
- Models favour labels frequent among the examples.

On the same model, how do prompting and fine-tuning compare for a multi-label task in a language with only projected labels?

Methodology

5941 Lines

in the Croatian data!

However...

XED / subtitle-retrieval / students / **pairs-hr.txt**

Code | Blame | 7679 lines (7679 loc) · 1

2363	en/1990/100507/4049754.xml.gz	hr			2079	Hit him, Dad. Udri ga, tata.
2364	en/1990/100507/4049754.xml.gz	hr			2087	You outclassed the bum, huh?
2365	en/1990/100507/4049754.xml.gz	hr			2090	Touch me and I'll sue. Pipni me
2366	en/1990/100507/4049754.xml.gz	hr			2092	Touch me and I'll sue. Pipni me
2367	en/1990/100507/4049754.xml.gz	hr			2097	- Come on, Speedy Gonzalez! - Haj
2368	en/1990/100507/4049754.xml.gz	hr			2106	This is where it all started for
2369	da/1954/46672/3631734.xml.gz	hr		4		De mener en kiste at ligge i! To znači,
2370	da/1954/46672/3631734.xml.gz	hr/1954/46672/5995891.xml.gz	17	8		Her er en, der var med på "Golden Arrow"
2371	da/1954/46672/3631734.xml.gz	hr/1954/46672/5995891.xml.gz	24	16 17		Vi blev torpederet hårdt om bagbord.
2372	da/1954/46672/3631734.xml.gz	hr/1954/46672/5995891.xml.gz	29	23		Det er et mirakel, at gamle Billy er i liv
2373	da/1954/46672/3631734.xml.gz	hr/1954/46672/5995891.xml.gz	33	27		Uh, den var hed som en ovn. Dah mu je kao
2374	da/1954/46672/3631734.xml.gz	hr/1954/46672/5995891.xml.gz	41	35 36		Kog hans sved ned, og der er gratis g

Only 2236 lines are directly (HR → EN) linked…

what do we do now?

Finding the English behind each Croatian line

Direct

2236 pairs

HR → EN

The Croatian line is linked straight to an English one.

Via Danish Linkage

3186 pairs

HR → DA → EN

XED only links it to a Danish subtitle, so we go through Danish, then link to English.

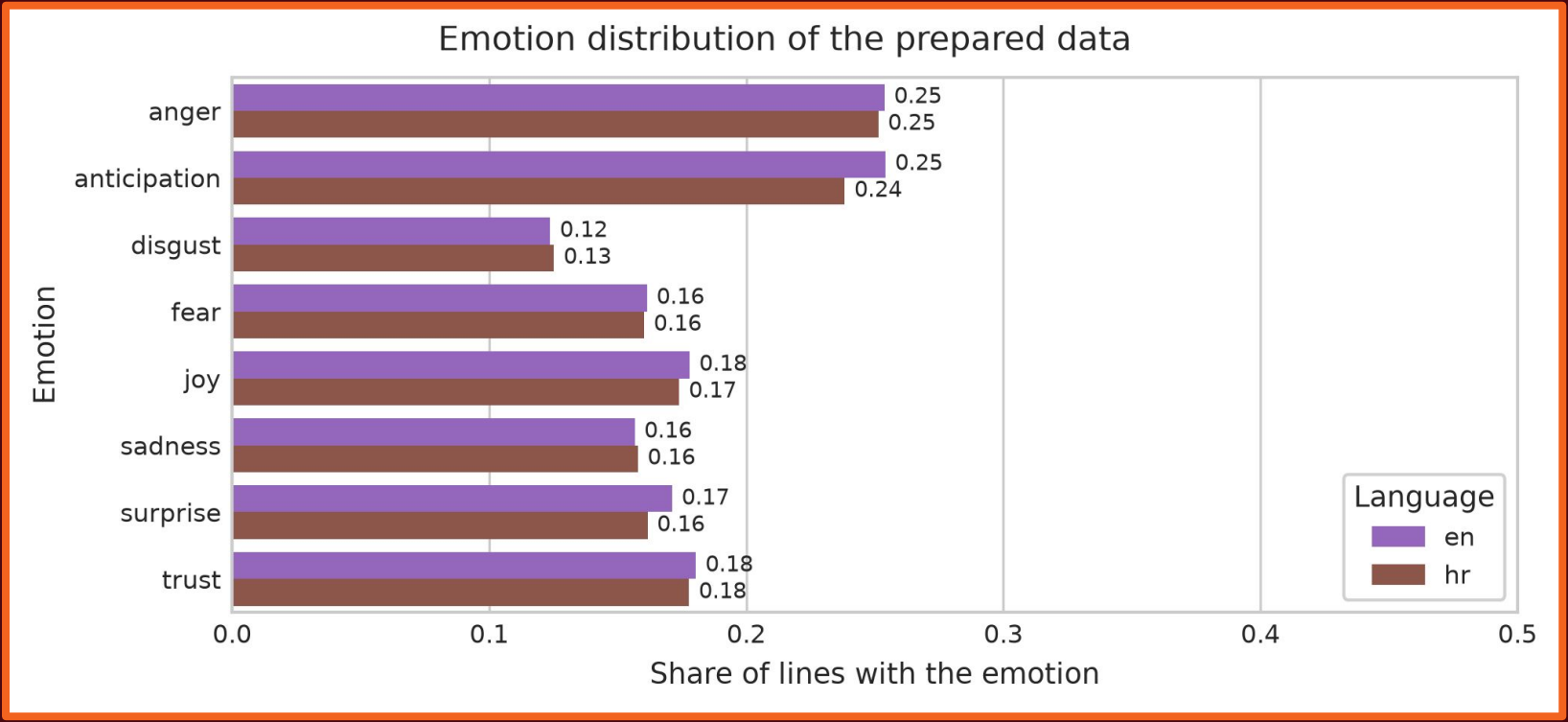
Result

5422 pairs

Malformed rows (1) and duplicates (518) were removed. Data split by subtitle, then stratified by label.

Train: 3803 (70%) | Dev: 1068 (20%) | Test: 551 (10%)

Emotion distribution in our languages



Main model: EuroLLM-9B

Why? Not an English-first model, built for all official EU languages.
Reference model: Qwen 3.5-9B (Prompt only); Croatian is undocumented; act as contrast.

Fine-tuning (QLoRA)

PEFT-en: English only.
PEFT-hr: Croatian only.
PEFT-Bilingual: Both.

Prompting

Same 4-bit model, weights unchanged.
Zero-shot: Instruction only.
Five-shot: Five solved examples.

Scoring

Same test pairs in both languages.
Micro-F1, Jaccard, macro-F1, precision, recall, Hamming loss, exact match, malformed-output rate.
95% bootstrap CIs, fine-tuning repeated with 3 seeds.

Implementation

Same input, same output

- The same prompt is used for prompting and fine-tuning.
- Croatian lines get the same English instruction.
- Fine-tuning learns to write the answer.
- Prompting writes it unaided.
- One parser reads every answer.

System
Label the subtitle with the emotions it expresses: anger, anticipation, disgust, fear, joy, sadness, surprise, trust (each with a short definition). Answer with a JSON list.

User
Subtitle: Znaš što?

Assistant
["surprise"]

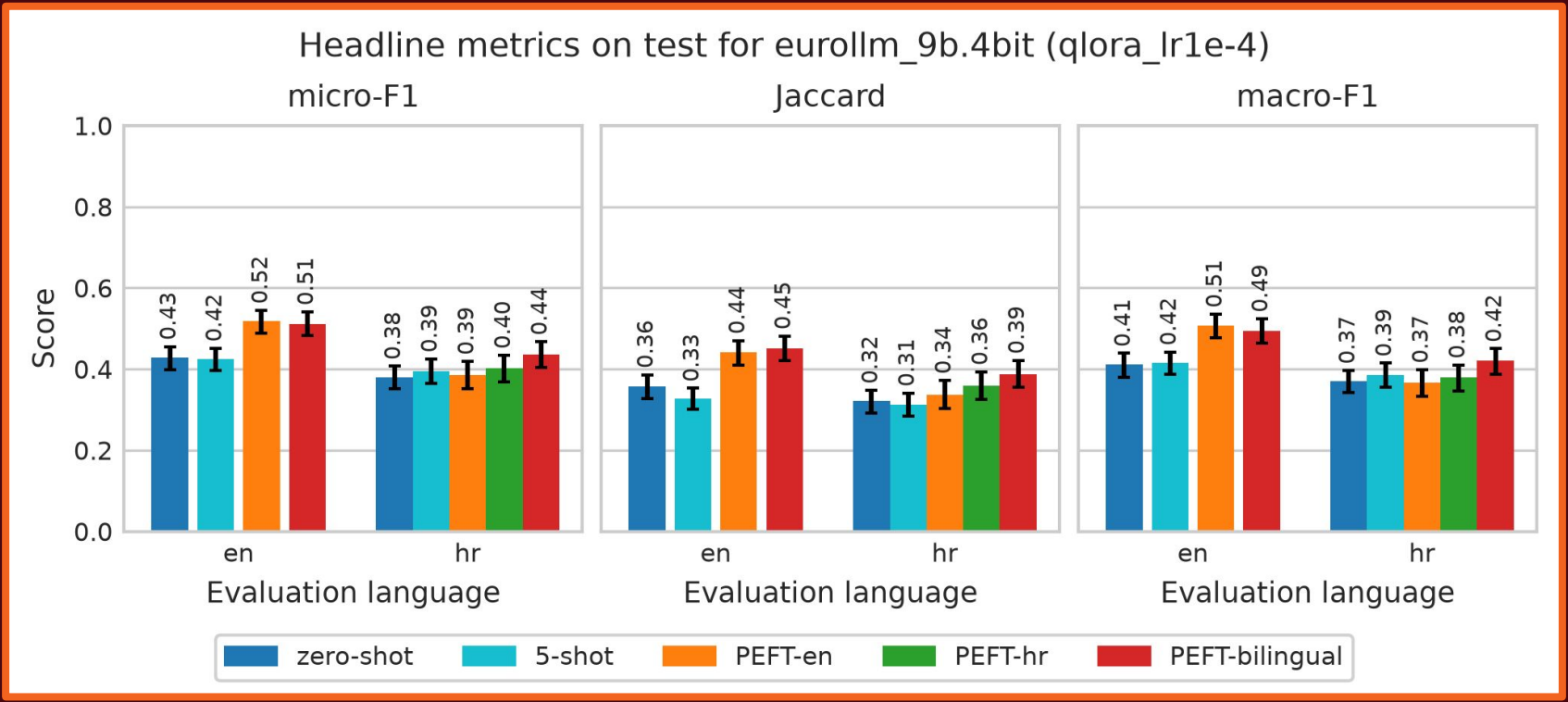
Our chosen five examples

Line	Label(s)
“Think I should go out with Paulie?”	Anticipation, surprise, trust
“Relax.”	Anger, fear, joy
“You got old.”	Disgust, sadness
“He said no pilots.”	Sadness, surprise.
“Shoot the fuck down!”	fear

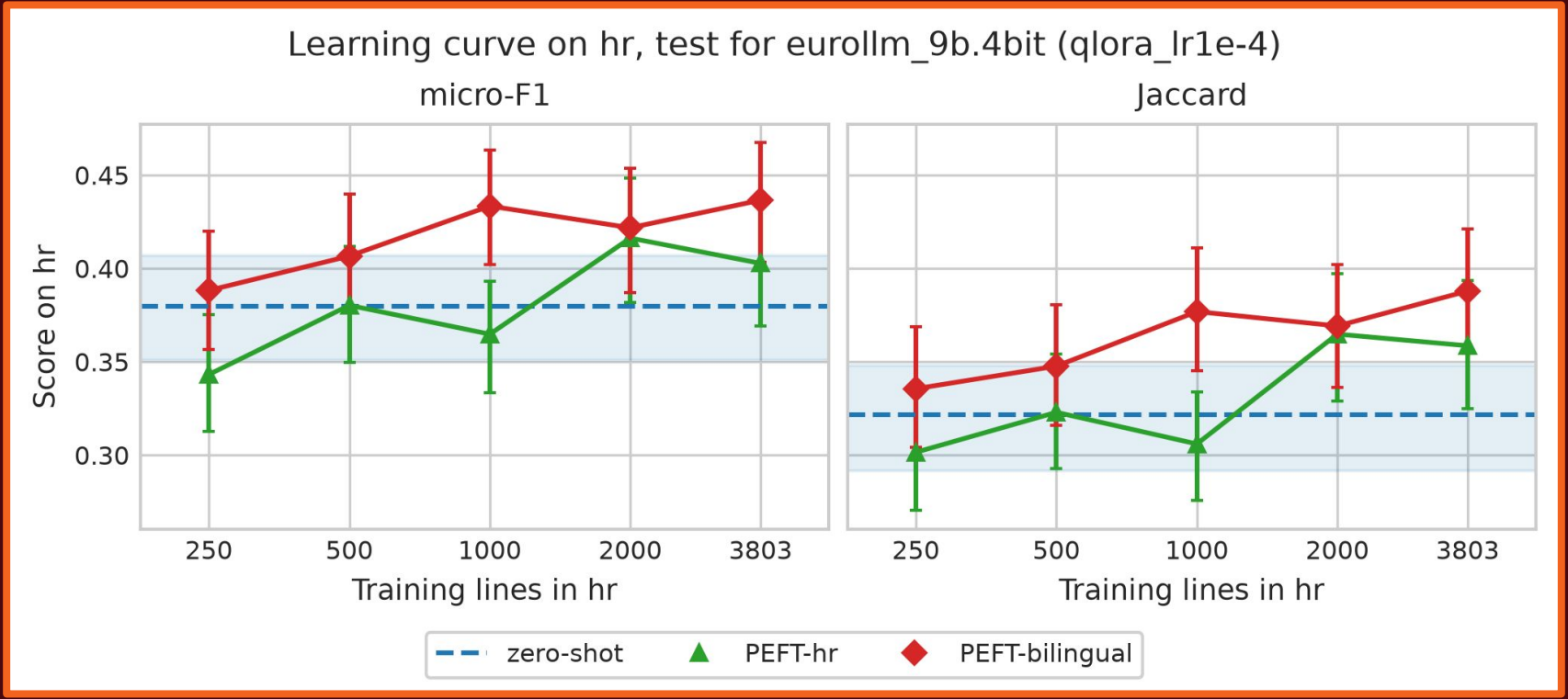
- Fixed in advance from the training split.
- Human English labels, direct Croatian match.
- All eight emotions covered.

Results & Discussion

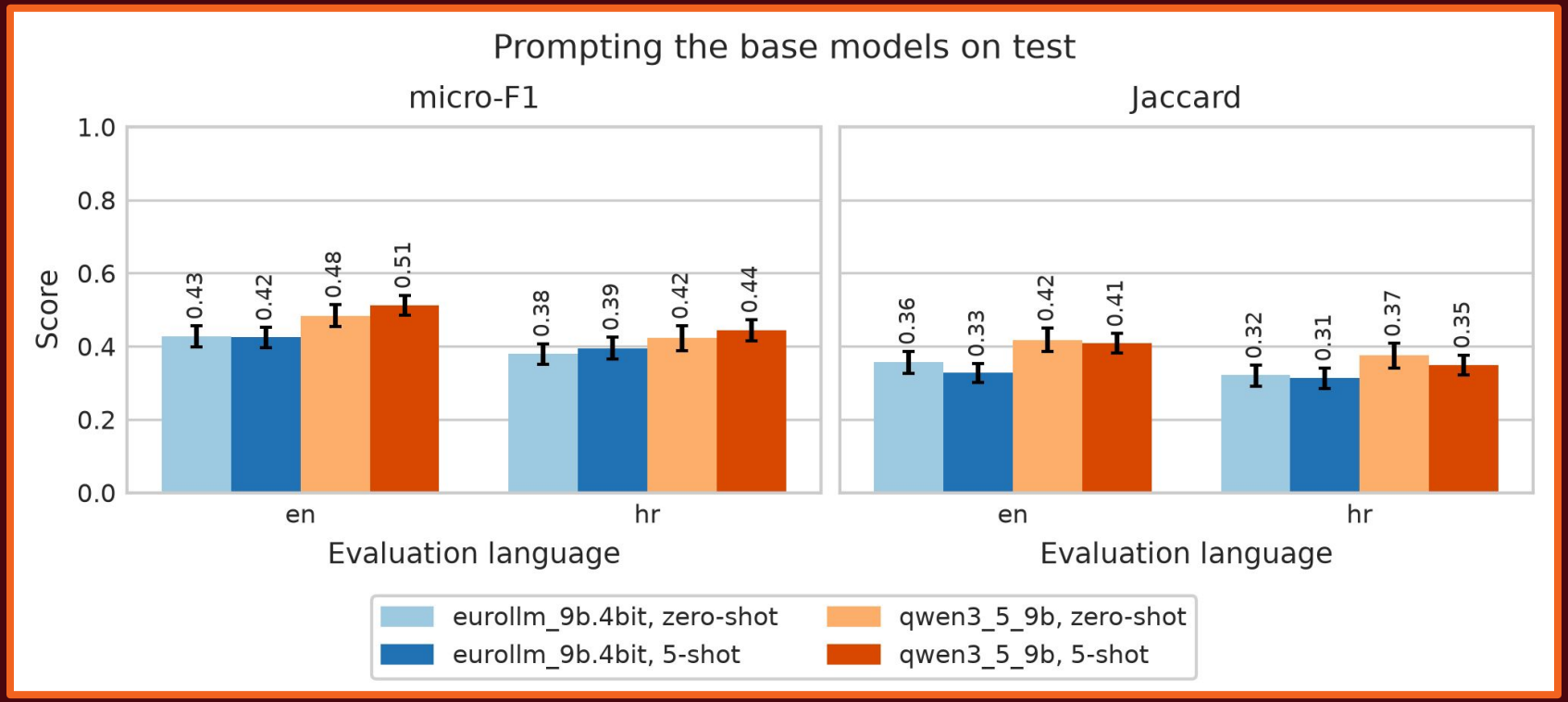
Fine-tuning wins while transfer falls short



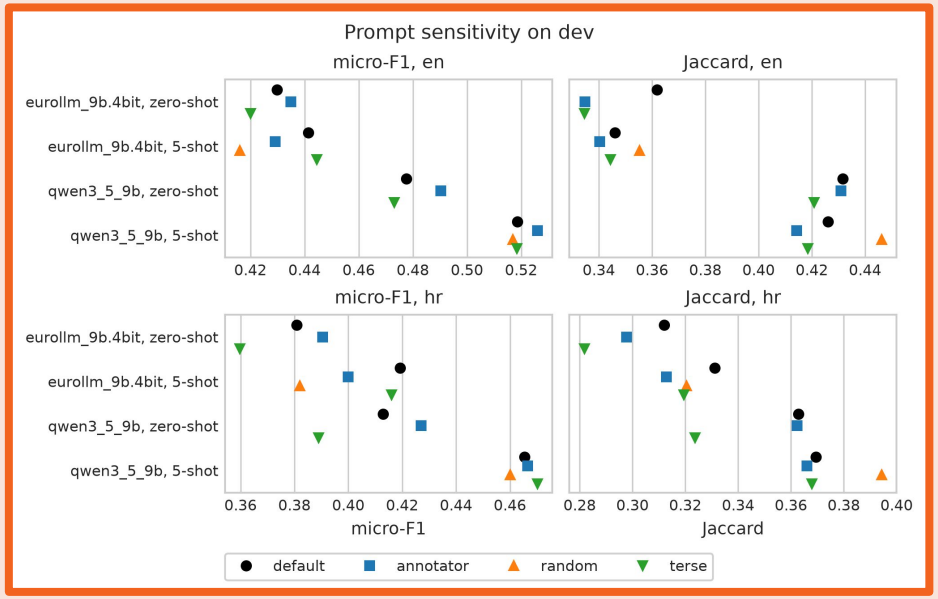
English helps, alongside Croatian



Qwen’s Croatian matches fine-tuned EuroLLM

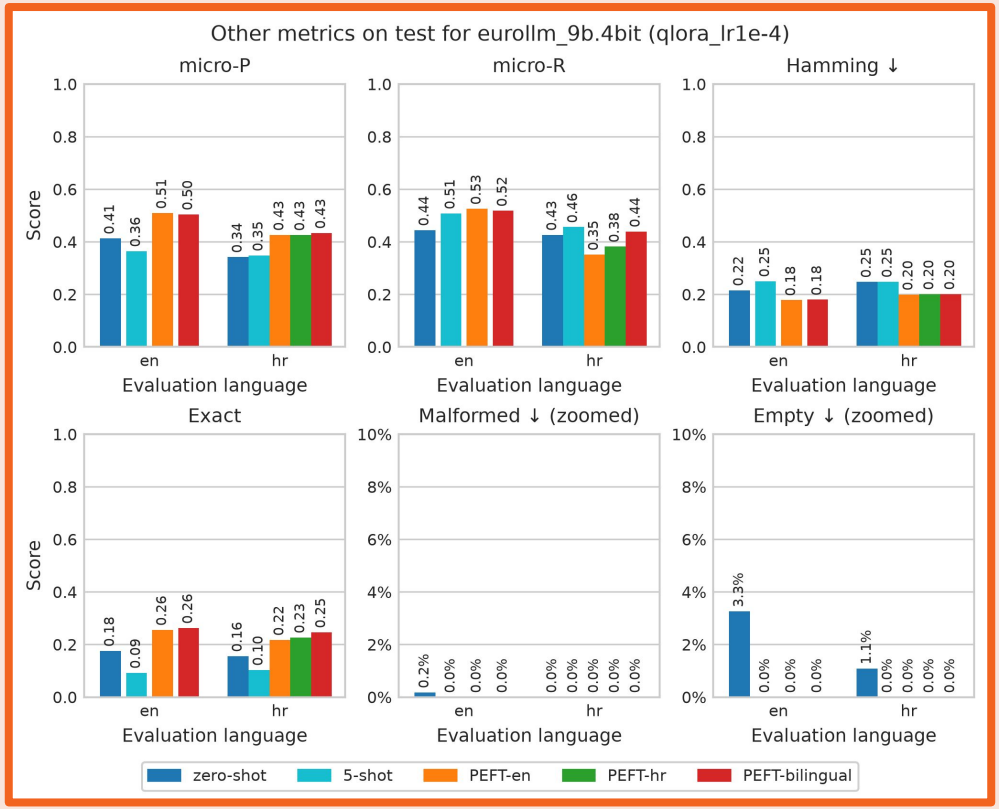


No prompt variant beats the default

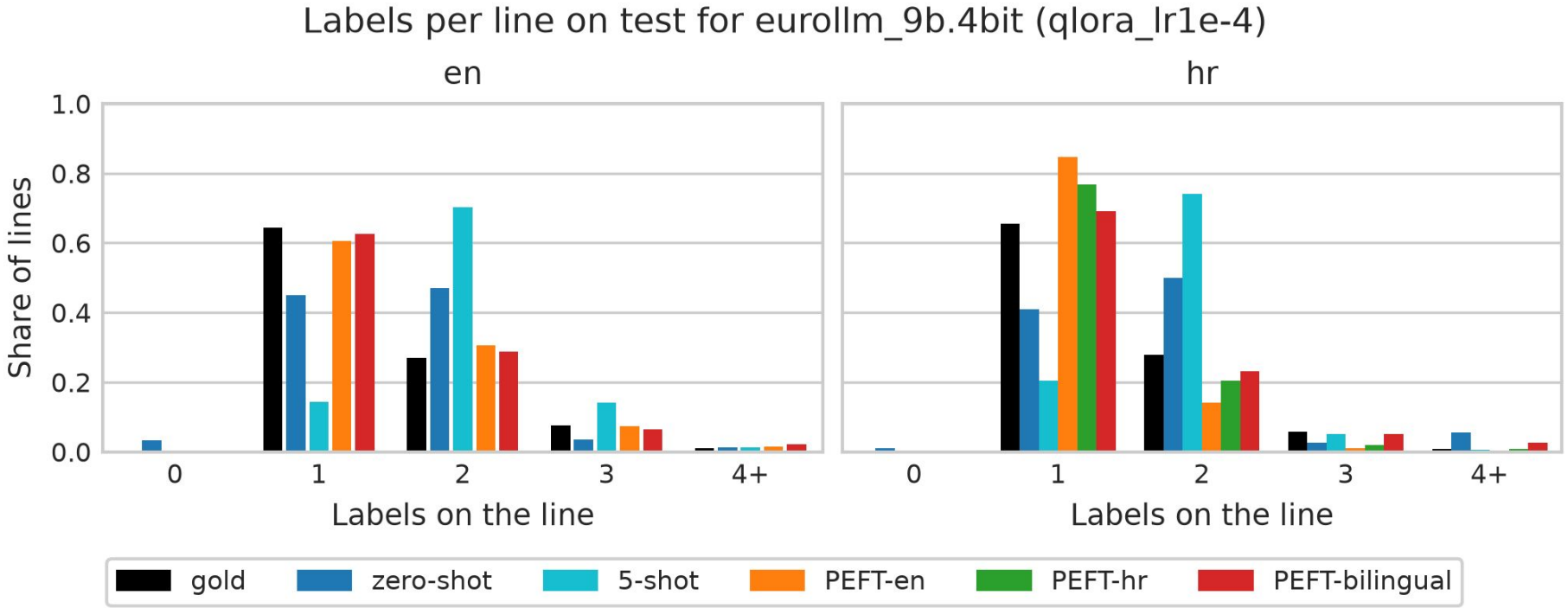


- **Rewording the instructions:** ≤ 0.04 dev micro-F1.
- Croatian is about twice as sensitive as English (zero-shot).
- **Chosen vs. random examples:** +0.025 en, +0.037 hr (EuroLLM).
- Default prompt within 0.014 of the best variant.

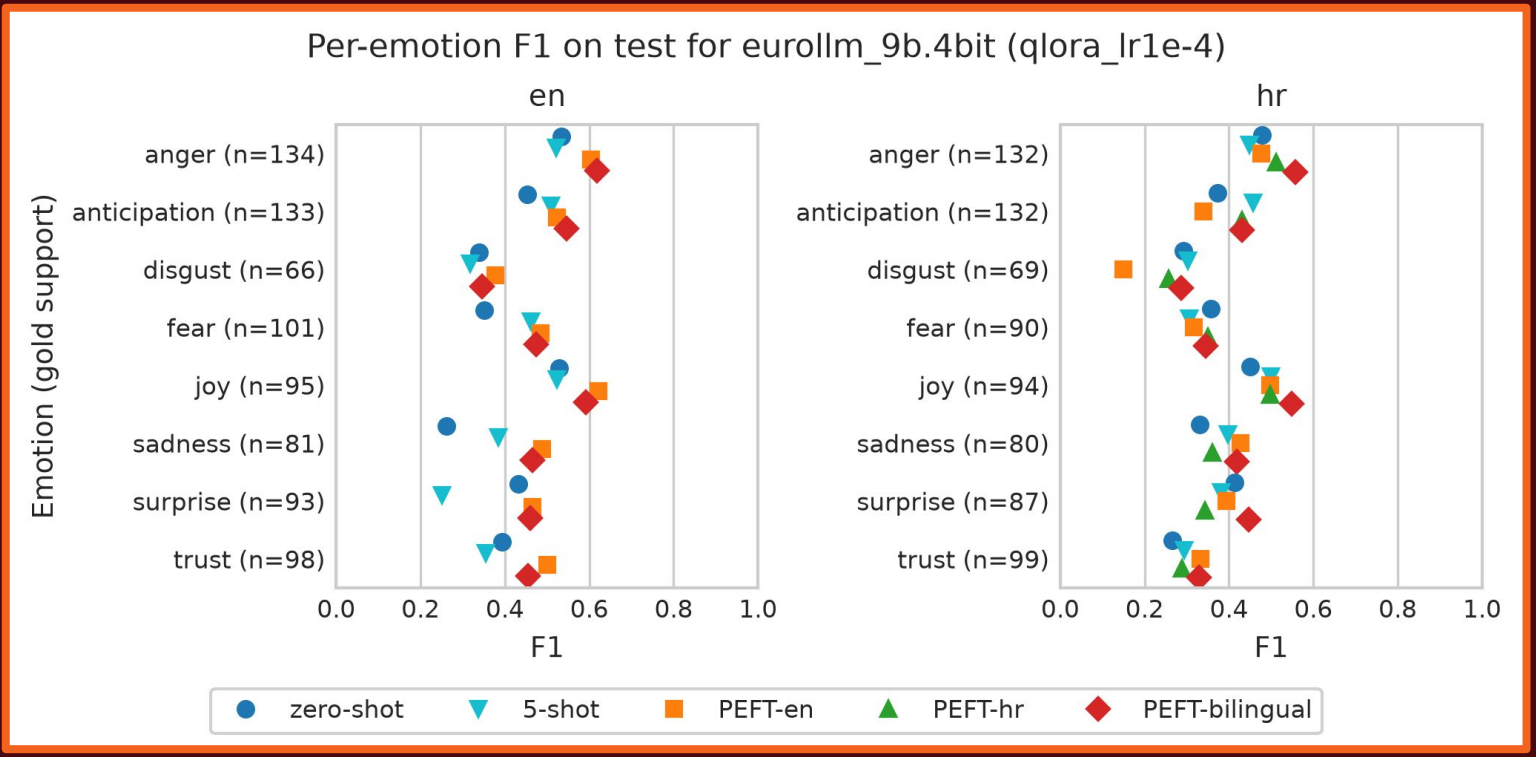
Beyond the primary metrics



Labelling really does matter



Disgust is hard for every system



Sometimes the label is wrong, not the model

“Što je to?”
(“What is that?”)

Aligned with: “Sign this.”

Inherited: anger, anticipation, joy, surprise

This is an English line, not a Croatian reader.

We do not speak Croatian, so we cannot verify.

“Žena je to jako teško prihvatila…”
(A woman took bad news hard)

Aligned with: “– What is it?”

Our model: sadness, scored as wrong

35/551 test labels disagree with their English source

Via Danish: 0.02-0.05 lower Jaccard, for every system.

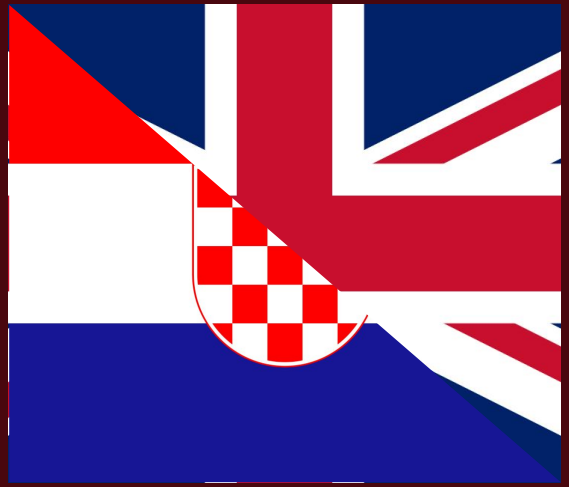
Conclusion

Back to our three questions...



Prompting or fine-tuning?

Fine-tuning; +0.09 micro-F1 in English.



Does English carry over?

Not alone; with approximately 1000 Croatian lines, it works best.



What do projected labels mean?

Part of the error is in the labelling, not the models.

Limitations

Labels:

- Croatian scores measure agreement with project labels, not with Natives.
- Approximately 175 English lines carry projected labels.
- Cannot separate weaker Croatian from label noise and make no claims about culture.

Scope:

- Only one language and fine-tuned model.
- Qwen was only prompted.
- No surrounding dialogue context.
- Test CIs of about ±0.03.
- PEFT-hr was still improving at the 3-epoch limit.



What could we do next?

Native review

- Verify the test labels
- Separate model errors from label errors.

Translate-test

- Classify machine-translated lines with PEFT-en.
- Determine if the Croatian gap is in the models or labels.

Filter misaligned pairs

- Keep pairs a sentence-embedding model calls “well aligned.”
- Project noise without a reader.

Widen the comparison

- Control label counts, add dialogue context.
- Fine-tune both models, compare resource bands.

Thank You!

Questions?

References

Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D., Wu, J., Winter, C., … Amodei, D. (2020). Language models are few-shot learners. In Advances in Neural Information Processing Systems (Vol. 33, pp. 1877–1901). Curran Associates, Inc. https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html

Dettmers, T., Pagnoni, A., Holtzman, A., & Zettlemoyer, L. (2023). QLoRA: Efficient finetuning of quantized LLMs. In Advances in Neural Information Processing Systems (Vol. 36, pp. 10088–10115). Curran Associates, Inc. https://doi.org/10.52202/075280-0441

Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., & Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In International Conference on Learning Representations. https://openreview.net/forum?id=nZeVKeeFYf9

Martins, P. H., Alves, J., Fernandes, P., Guerreiro, N. M., Rei, R., Farajian, A., Klimaszewski, M., Alves, D. M., Pombal, J., Boizard, N., Faysse, M., Colombo, P., Yvon, F., Haddow, B., de Souza, J. G. C., Birch, A., & Martins, A. F. T. (2025). EuroLLM-9B: Technical report. arXiv. https://doi.org/10.48550/arXiv.2506.04079

Öhman, E., Pàmies, M., Kajava, K., & Tiedemann, J. (2020). XED: A multilingual dataset for sentiment analysis and emotion detection. In Proceedings of the 28th International Conference on Computational Linguistics (pp. 6542–6552). International Committee on Computational Linguistics. https://doi.org/10.18653/v1/2020.coling-main.575

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

Qwen Team. (2026, February). Qwen3.5: Towards native multimodal agents. https://qwen.ai/blog?id=qwen3.5

Sechidis, K., Tsoumakas, G., & Vlahavas, I. (2011). On the stratification of multi-label data. In Machine Learning and Knowledge Discovery in Databases (pp. 145–158). Springer. https://doi.org/10.1007/978-3-642-23808-6_10

Zhao, Z., Wallace, E., Feng, S., Klein, D., & Singh, S. (2021). Calibrate before use: Improving few-shot performance of language models. In Proceedings of the 38th International Conference on Machine Learning (Vol. 139, pp. 12697–12706). PMLR. https://proceedings.mlr.press/v139/zhao21c.html

Appendix

Prompt used on test

System Message (Default)

You classify the emotions expressed in a movie subtitle line.
Permitted labels:
{definitions}
Every line expresses at least one of these emotions and may express several. Answer with a JSON list of the matching label names, using only the names above, and nothing else, in the format ["label", ...].

User Message

Subtitle: {text}

–

Five-shot: Five demonstrations go in a user and assistant turns between the two.

Decoding: Greedy, at most 32 new tokens.

Qwen’s reasoning mode turned off.

Label Definitions {{definitions}}

anger: displeasure or hostility at a perceived wrong
anticipation: expecting or looking forward to something
disgust: revulsion or strong disapproval
fear: feeling threatened or afraid
joy: happiness or pleasure
sadness: sorrow, loss or disappointment
surprise: reaction to something unexpected
trust: confidence in, or acceptance of, someone or something

Prompt Variations

Template “terse”

Label the emotions in this subtitle line. Choose one or more of: {definitions} Reply with only a JSON list of the chosen label names, like ["label", ...].

Template “annotator”

You are an expert annotator of emotions in film dialogue, using Plutchik's eight basic emotions: {definitions} Read the subtitle line and decide which of these emotions it expresses; a line can express one emotion or several at once. Respond with a JSON list of the label names exactly as written above, and no other text.

Demonstrations “random”

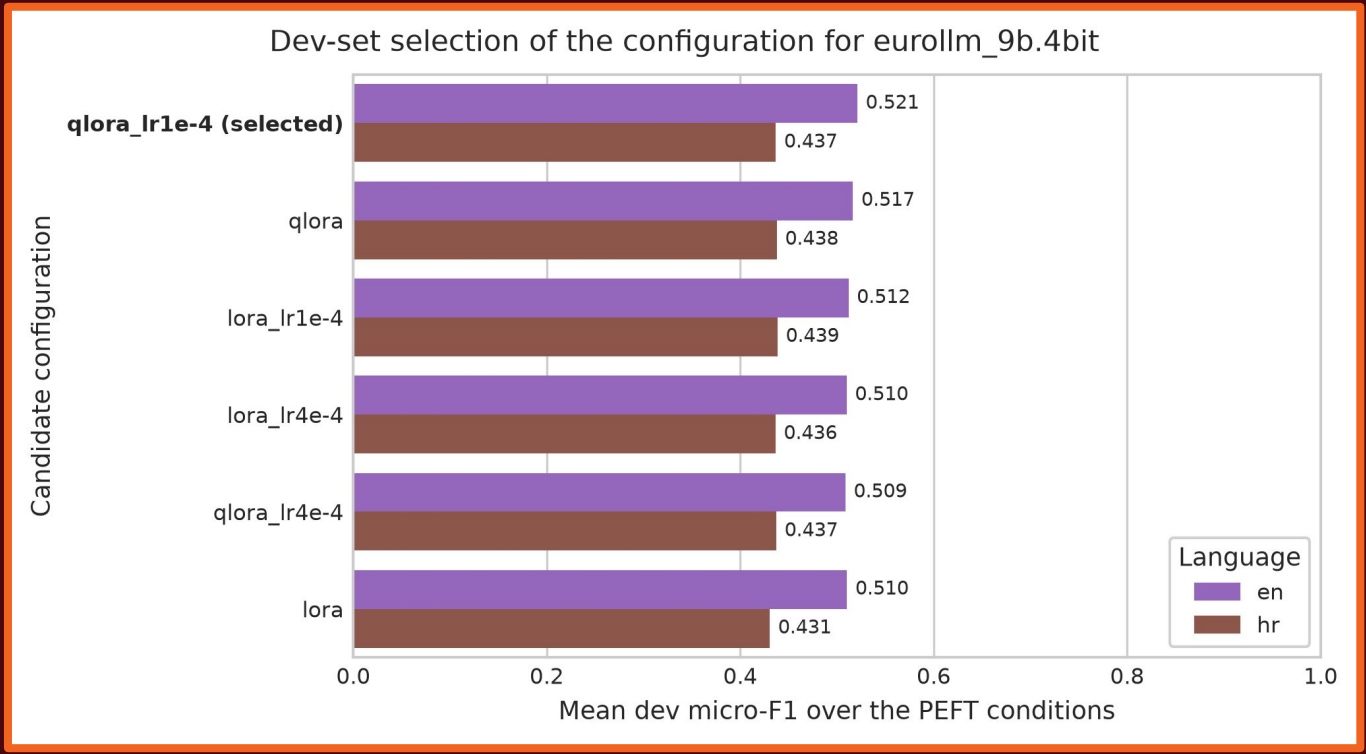
Five random training lines instead of the five emotion-covering ones.

The five fixed demonstrations

#	English	Croatian	Labels (en)	Labels (hr)
1	- Think I should go out with Paulie?	- Hoćeš da izađem sa Paulijem?	anticipation, surprise, trust	anticipation, surprise, trust
2	Relax.	Smiri se.	anger, fear, joy	anger, fear
3	You got old.	Ostarili ste.	disgust, sadness	disgust, sadness
4	He said no pilots.	Neće pilota.	sadness, surprise	sadness, surprise
5	Shoot the fuck down!	Skini ih jebeno dolje!	fear	fear

Chosen from the training split: human English labels, a direct Croatian alignment, at most 80 characters and 3 labels.
Greedily picked to cover the most emotions, one per film (seed 42).
About two labels per line, against approximately 1.45 in the data.

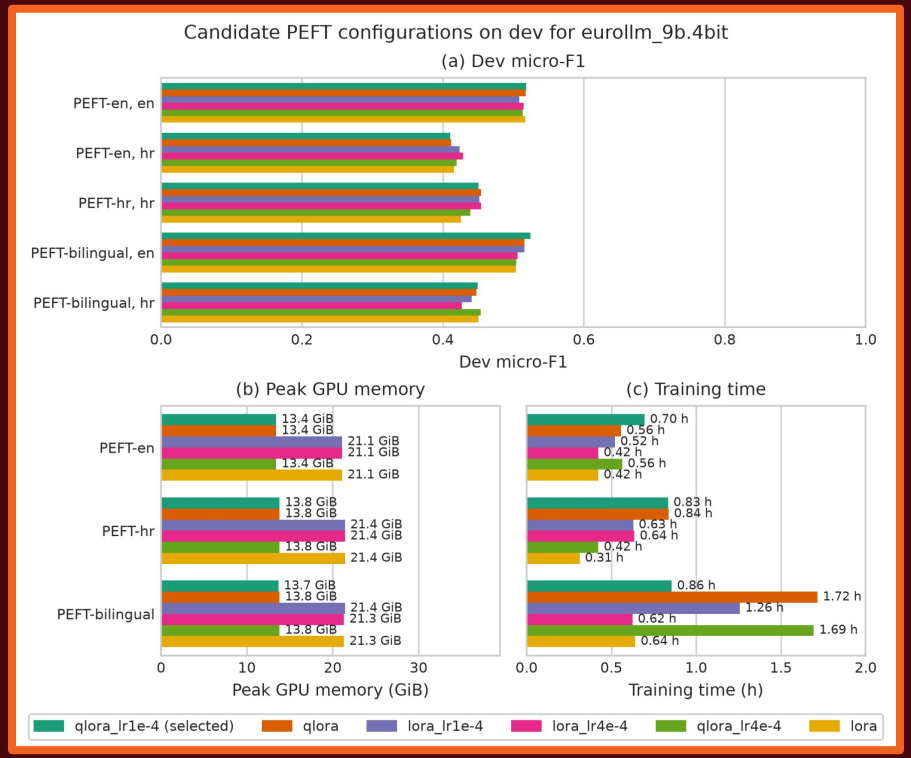
What was the final candidate for dev?



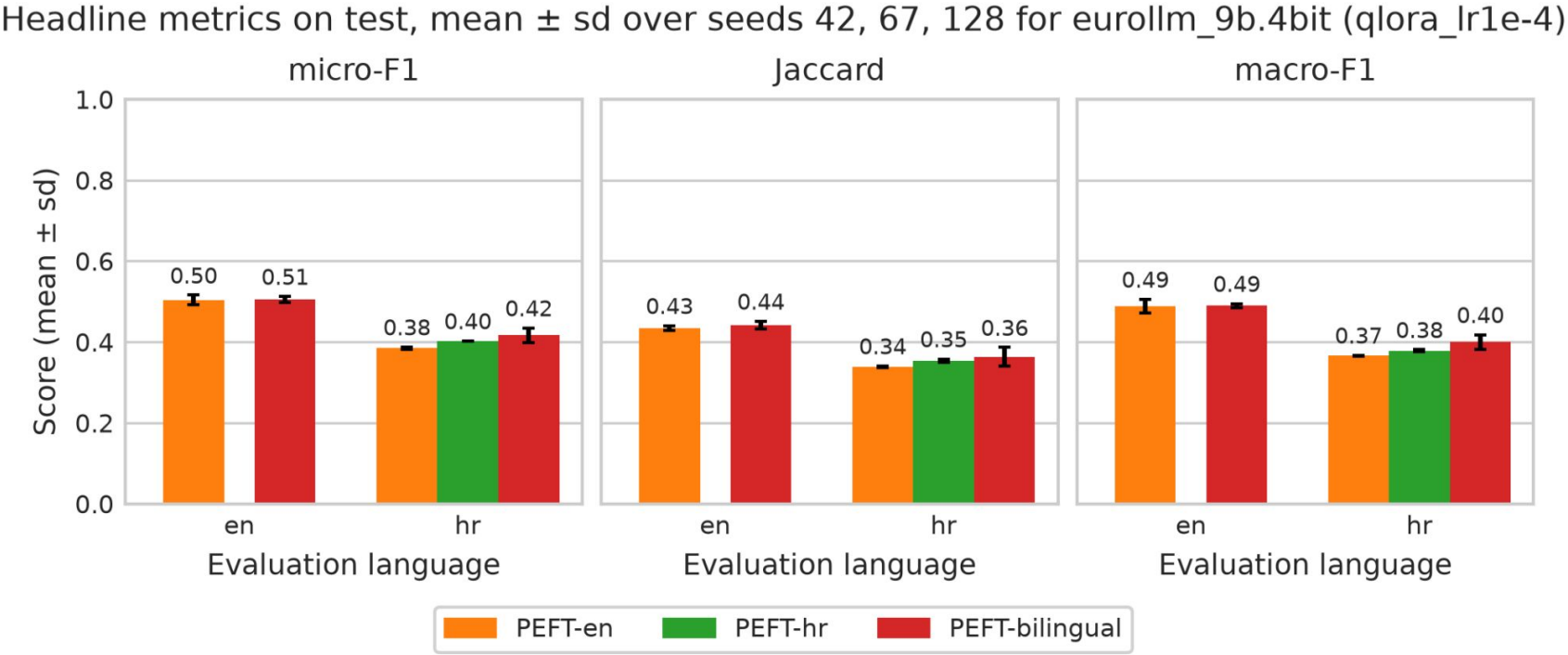
Fine-tuning configuration

Setting	Value
Base model	EuroLLM-9B-Instruct (rev. f7ae2bc3), 4-bit NF4 with double quantisation.
Adapters	QLoRA: LoRA adapters on all linear layers; r = 16, α = 32, dropout 0.05 · 50.9M trainable (0.55%).
Optimiser	AdamW, no weight decay; lr 1×10^{-4}, 3% warm-up, linear decay.
Batch and length	8 × 2 gradient accumulation = 16; max 256 tokens.
Training	≤ 3 epochs; seed 42; deterministic; gradient checkpointing.
Checkpoint rule	Dev micro-F1 twice per epoch; stop after 2 evaluations without gain.
Checkpoints kept	PEFT-en epoch 1.5; PEFT-hr 3.0; PEFT-bilingual 0.5.
Cost per run	0.70–0.86 h; ≤ 13.8 GiB GPU memory.
Hardware	1× NVIDIA A100 40 GB, university HPC cluster.
Software	PyTorch 2.14.0; Transformers 5.17.0; PEFT 0.21.2; scikit-learn 1.9.1.

Configuration search on dev

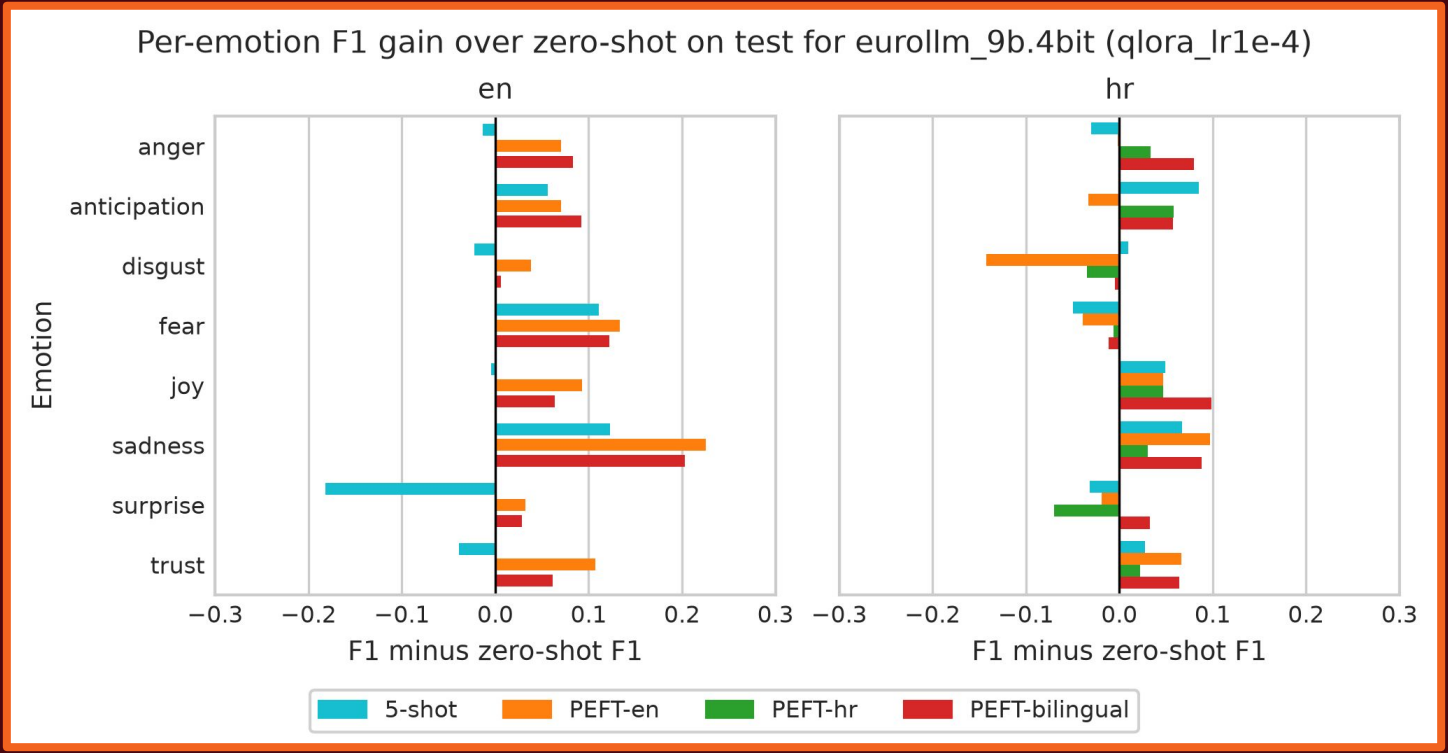


Exploration on three seeds



System	Micro-F1 [95% CI]	Jaccard [95% CI]	macro-F1	P	R	HL	EM
English							
EuroLLM zero-shot	0.428 [.398, .456]	0.357 [.327, .386]	0.412	0.413	0.444	0.26	0.176
EuroLLM five-shot	0.425 [.396, .451]	0.327 [.301, .353]	0.415	0.365	0.508	0.250	0.093
PEFT-en	0.517 [.488, .545]	0.441 [.410, .470]	0.508	0.510	0.526	0.178	0.256
PEFT-bilingual	0.511 [.483, .541]	0.451 [.420, .481]	0.494	0.504	0.518	0.180	0.263
Qwen zero-shot	0.484 [.454, .513]	0.417 [.386, .450]	0.459	0.540	0.438	0.170	0.263
Qwen five-shot	0.513 [.485, .539]	0.408 [.381, .435]	0.505	0.447	0.602	0.208	0.143
Croatian							
EuroLLM zero-shot	0.380 [.351, .407]	0.322 [.292, .348]	0.370	0.342	0.427	0.247	0.156
EuroLLM five-shot	0.395 [.365, .425]	0.312 [.285, .341]	0.386	0.348	0.456	0.248	0.103
PEFT-en	0.385 [.351, .419]	0.337 [.303, .372]	0.367	0.426	0.351	0.199	0.218
PEFT-hr	0.403 [.369, .434]	0.359 [.325, .393]	0.380	0.426	0.382	0.201	0.227
PEFT-bilingual	0.436 [.403, .467]	0.388 [.355, .421]	0.420	0.434	0.439	0.202	0.247
Qwen zero-shot	0.422 [.389, .456]	0.375 [.341, .408]	0.408	0.471	0.383	0.186	0.260
Qwen five-shot	0.444 [.414, .473]	0.349 [.322, .376]	0.438	0.386	0.524	0.233	0.120

Per-emotion F1 gain over zero-shot



Labels via Danish score a little lower

